

CAS Machine Learning – Time Series

Conclusion

Dr. Ludovic Amruthalingam
ludovic.amruthalingam@hslu.ch

Informatik



Outline

Common mistakes in time series analysis

- Data analysis
- Splitting
- Evaluation workflow

Reading ML research papers

- Why?
- How?

Common mistakes – Data analysis

Ignoring **data quality** issues

- Missing values, outliers, and inconsistent timestamps.
- Data format mismatches (e.g., date-time parsing errors).

Insufficient exploratory analysis

- **No visualization**; missing key patterns, seasonality, or trends.
- Skipping basic summary statistics, leading to poor data understanding.

Data transformations

- Applying smoothing, differencing, or normalization without **considering their impact** on data structure.
- Using transformations that **introduce artifacts or distort** the underlying signal.

Lack of **domain knowledge**

- Over-reliance on AutoML e.g., automated feature extraction.
- Neglecting known external factors (e.g., holidays, policy changes, market shocks).

Best practices for data analysis

Remember the **garbage in, garbage out** principle.

Documentation: keep records of every data cleaning/transformation steps for reproducibility.

Perform **sanity checks** on the raw data, after pre-processing and after transformations.

Visualize more than less: use variety of plots (line charts, seasonal plots, ACF plots) to gain insights.

Contextualize the data

- Understand the data generation process.
- Interpret outliers, relate to real-world events.
- Ensure transformations and assumptions make sense.
- Interpret results with domain knowledge.

Common mistakes – Splitting

Messing the **temporal order**

- Using random splits as if samples were independent and identically distributed.
- Forecasting using future observations.

Splitting the dataset without considering **seasonal** cycles or **trend** patterns.

Selecting too short test period that doesn't capture **long-term patterns**.

Not considering **regime changes** in the data.

Data splits with **different distribution**.

Common mistakes – Evaluation workflow

Not reserving test data: performance evaluation on **data seen during training**.

Overfitting to the validation set by repeatedly adjusting parameters without a separate hold-out test set.

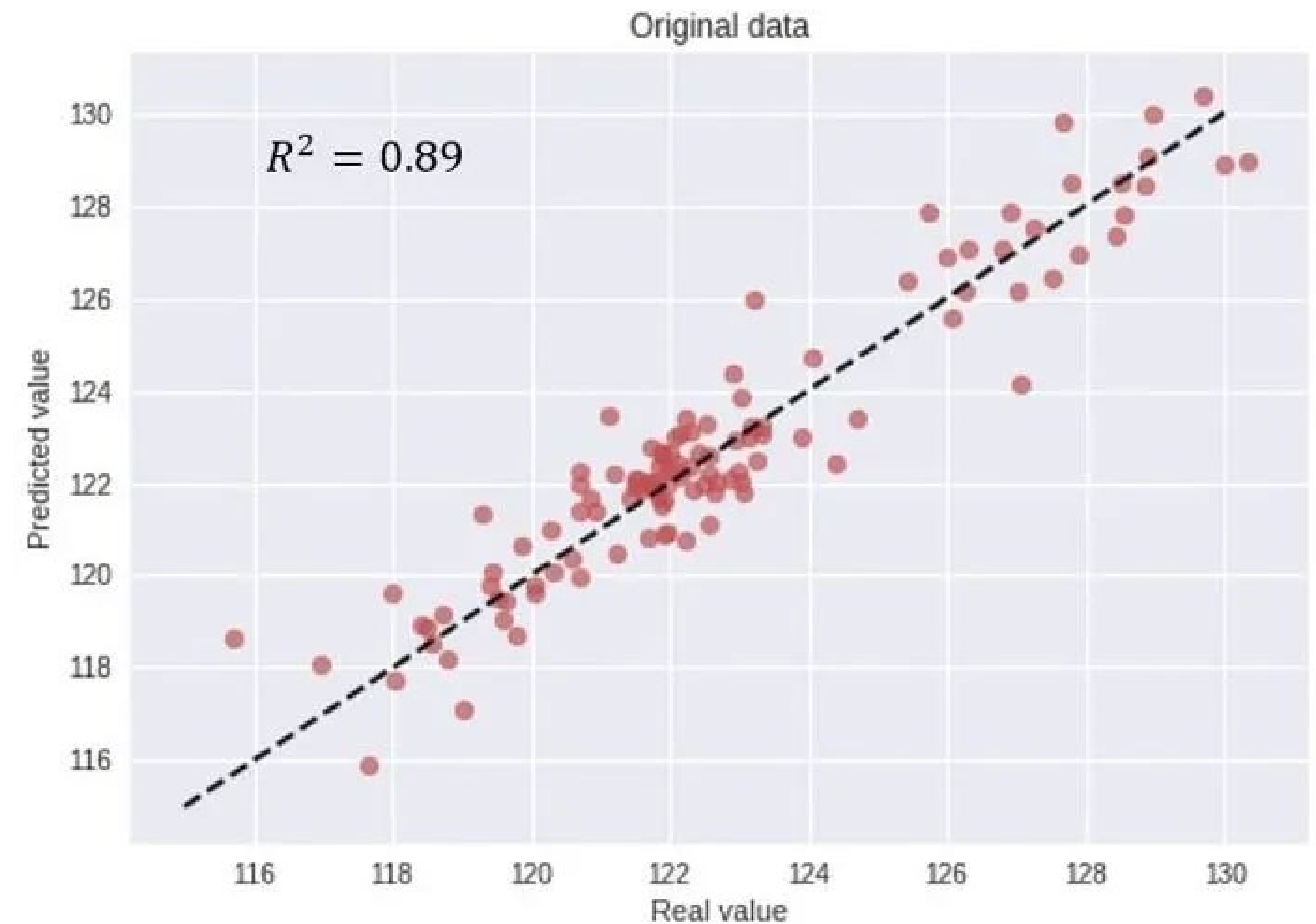
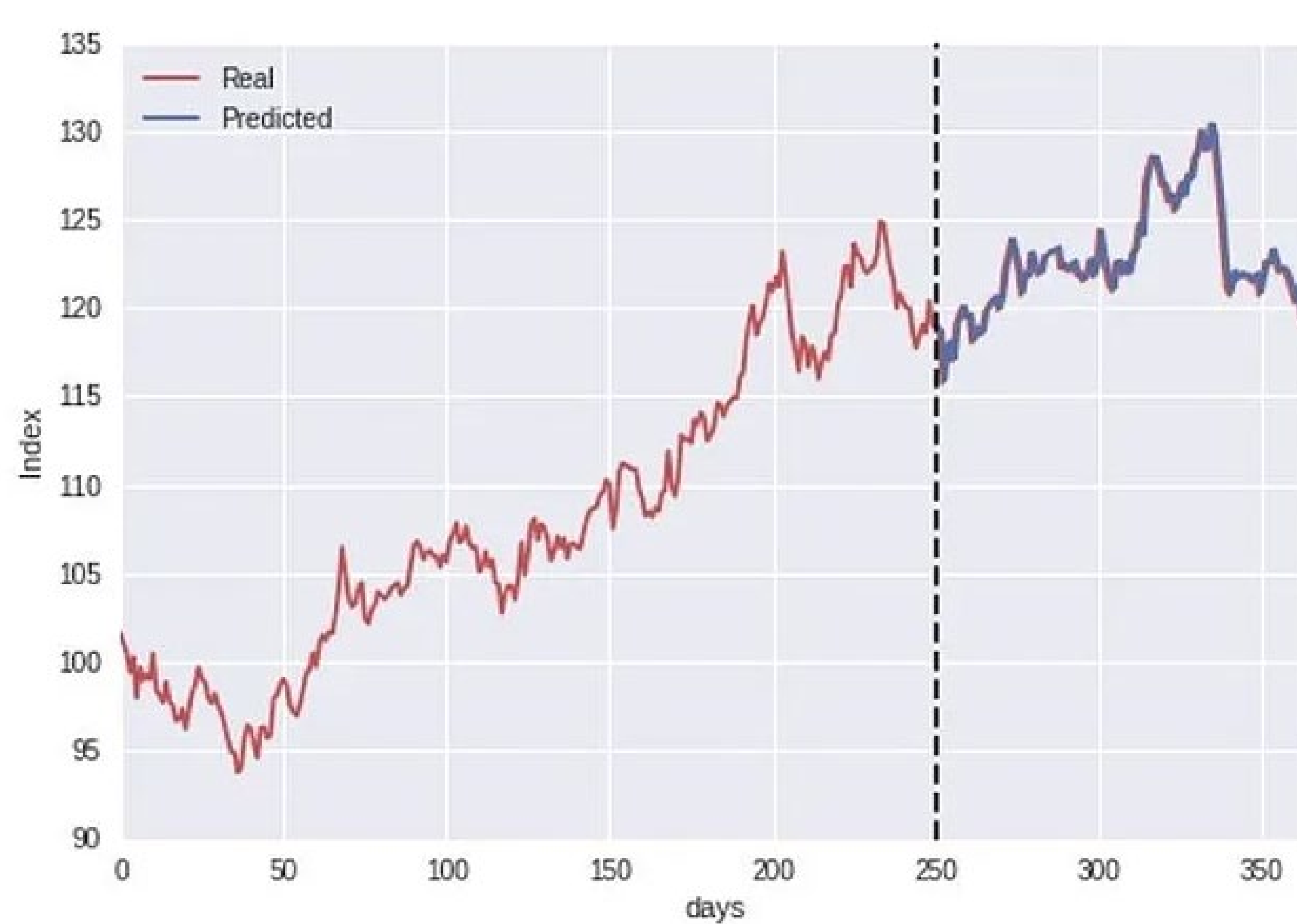
Not comparing against simple **baselines**, using inappropriate performance metrics.

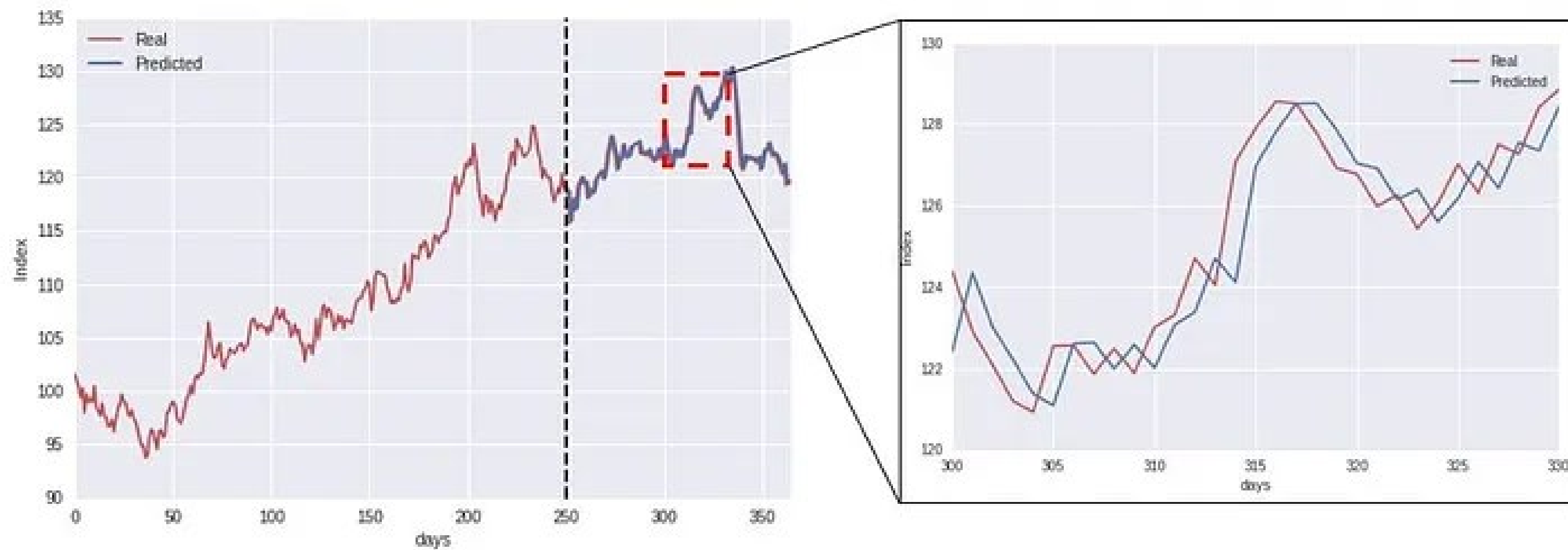
No quantification of **uncertainty**.

Using a single validation window instead of multiple rolling or expanding windows.

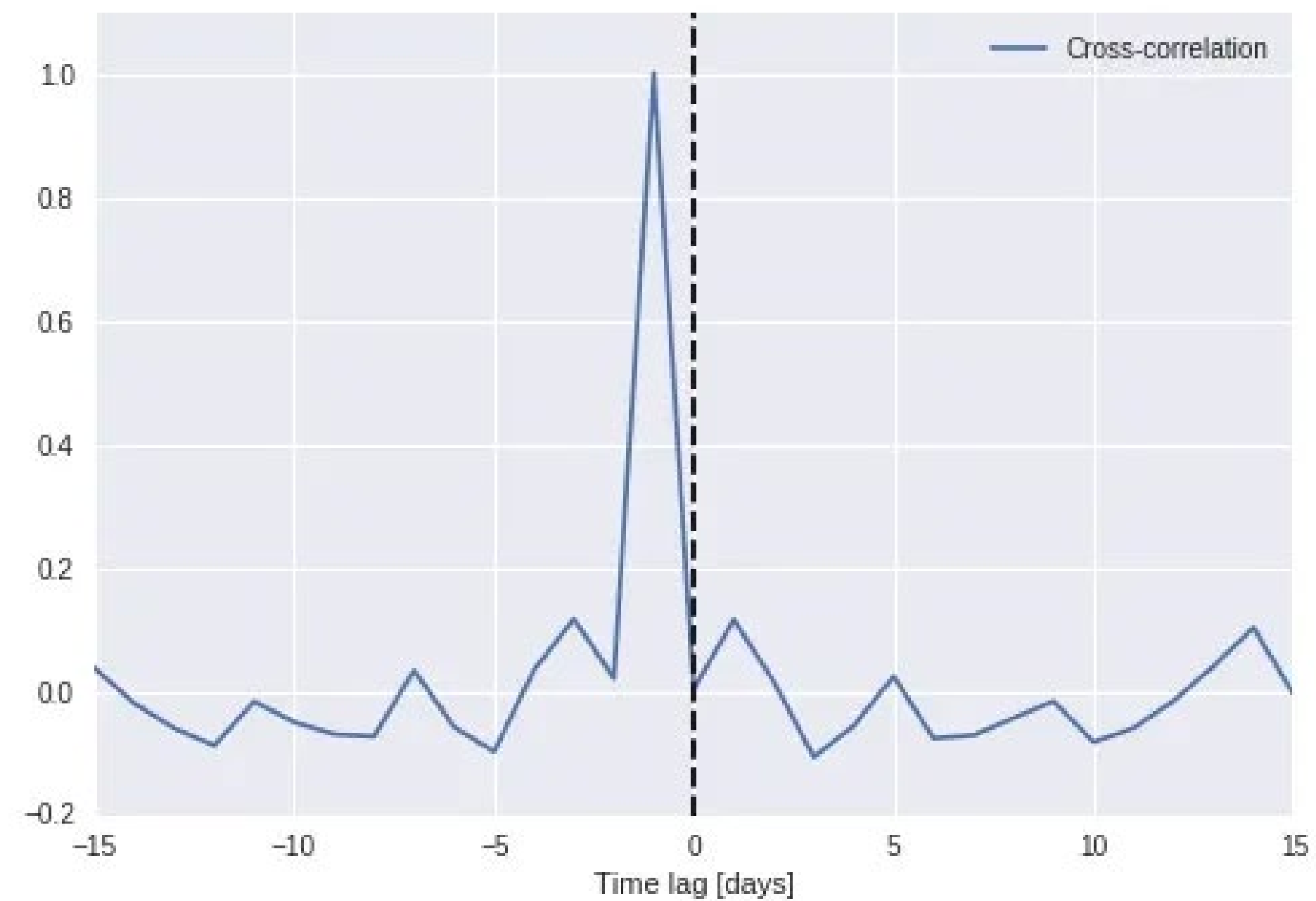
Not evaluating models on **out-of-distribution** samples e.g., data collected in different centers.

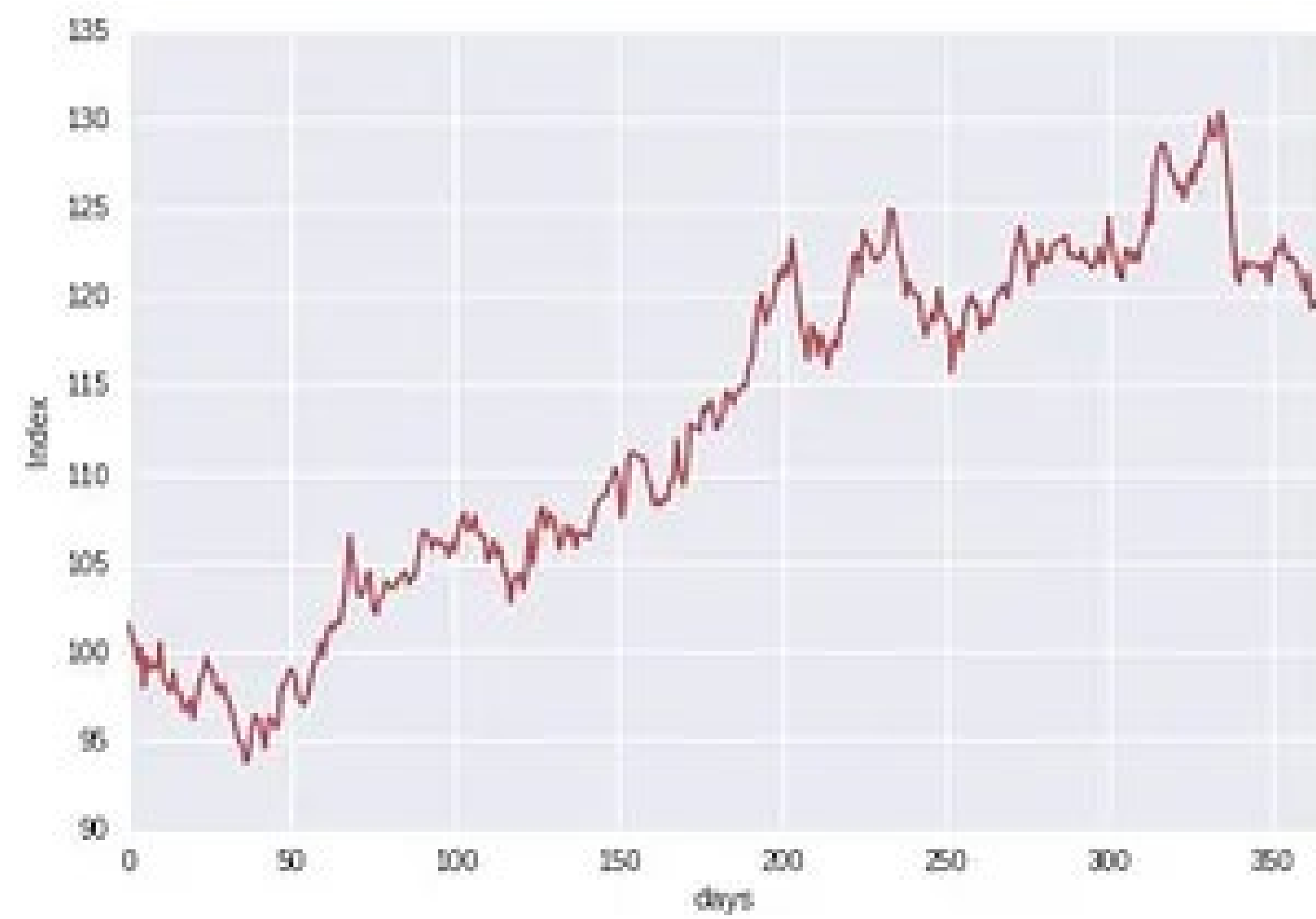
When performance are too good to be true...



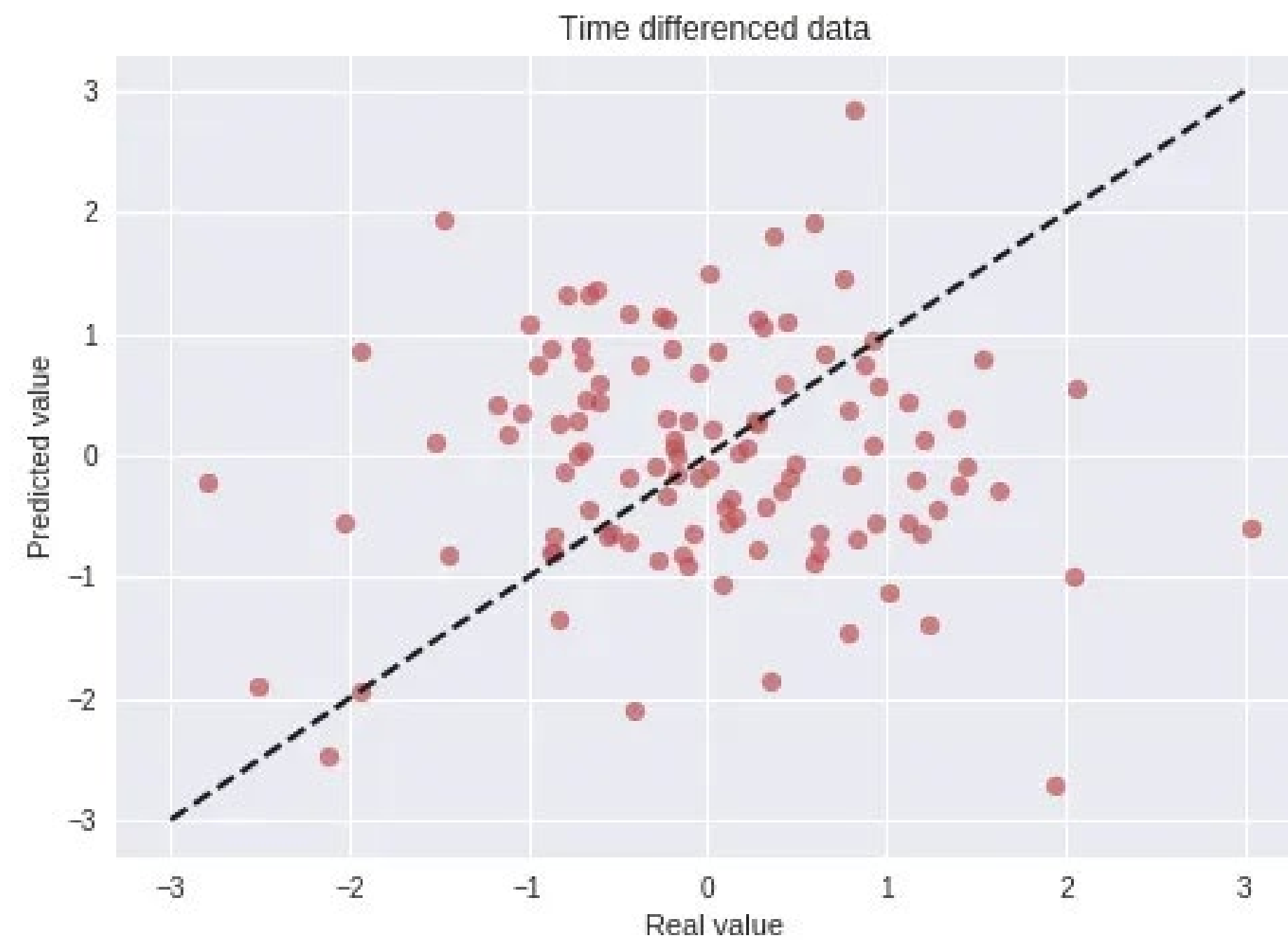
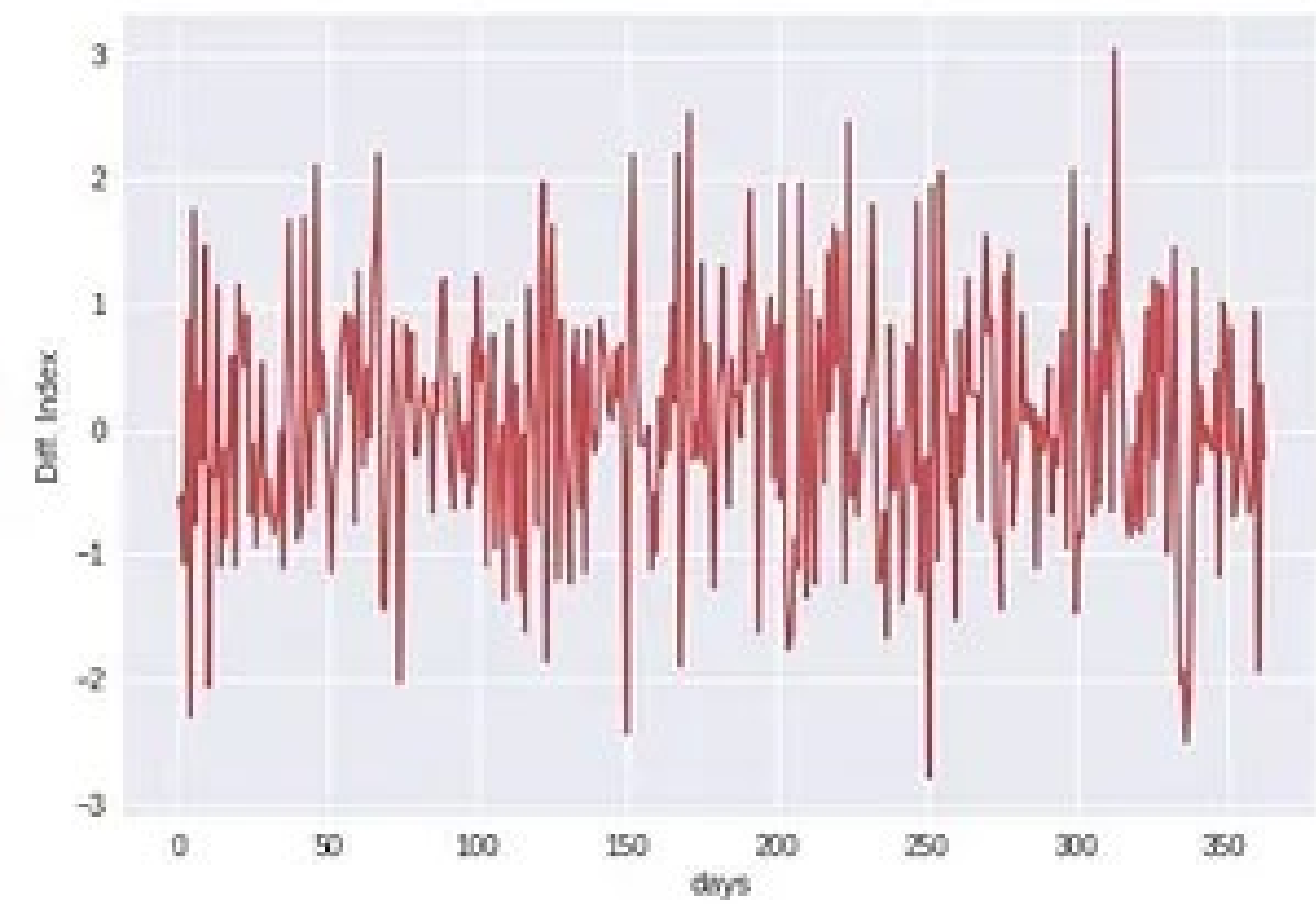


Cross-correlation between predicted and real value

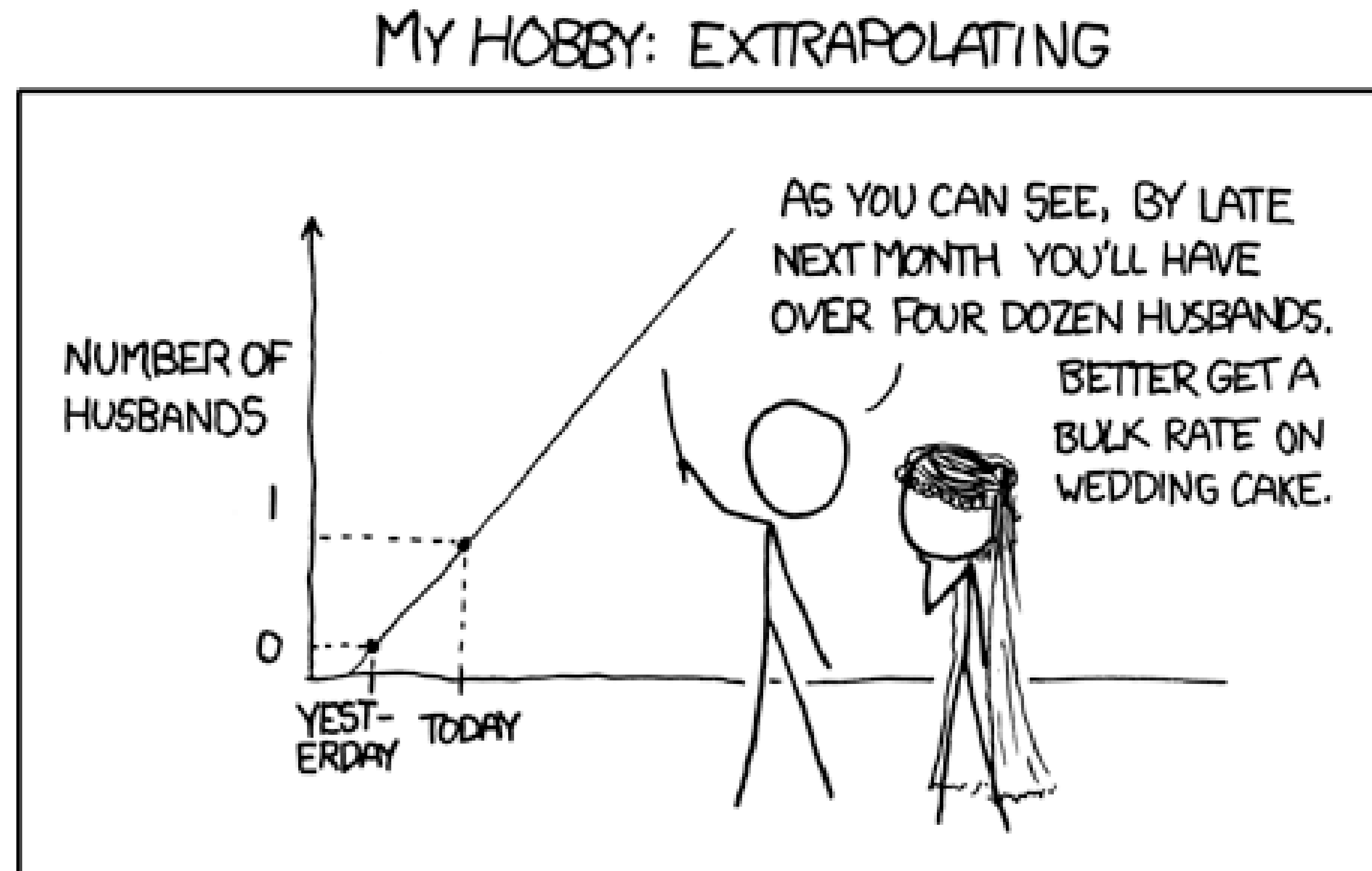




Time differencing



Common mistake – Extrapolating linear regression



Randall Munroe, <https://xkcd.com/605/>

Time is infinite and **ever-increasing**.

Linear regression with time as independent variable should **not** be extrapolated beyond **short time scale**.

Common mistakes

Aliasing due to inappropriate sampling rate.

Granularity of timestamps: too coarse/fine can obscure patterns.

Misalignment between **forecast horizon** and actual needs.

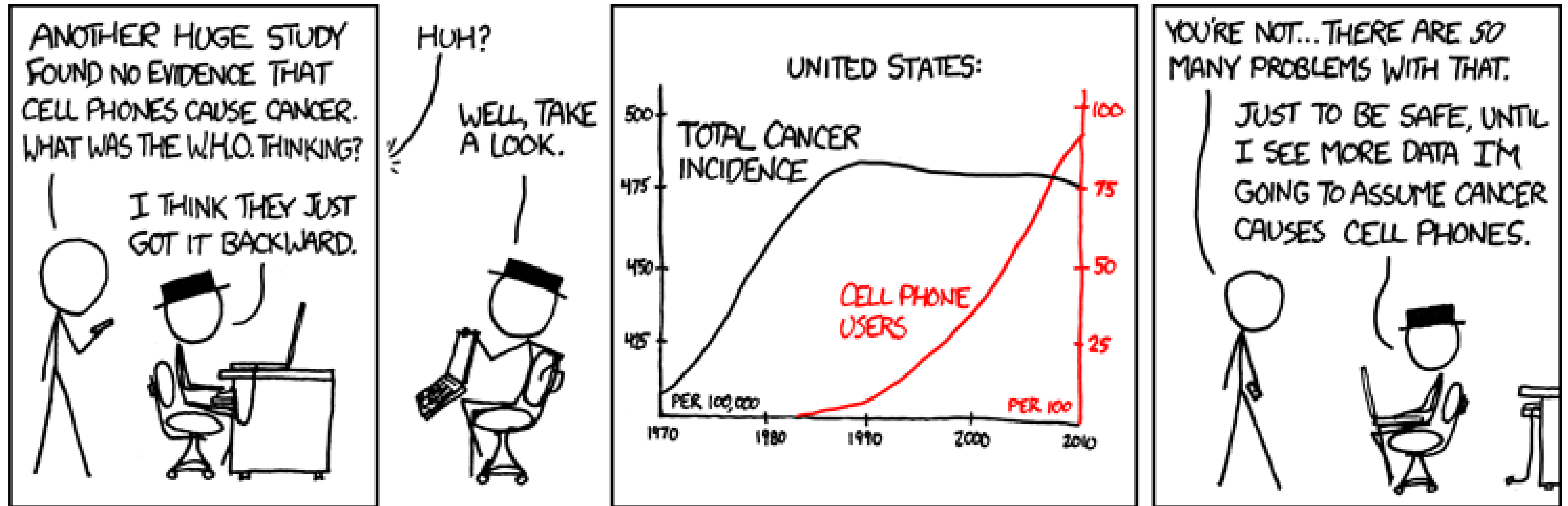
Model **assumptions** and uncertainty evaluation: stationarity, error distribution etc.

Spurious correlations: mistaking random fluctuations for patterns.

Data spanning too short period of time may hide underlying patterns.

Failure to detect/model **regime changes**.

Correlation/causation fallacy



Randall Munroe, <https://xkcd.com/925/>

ML research papers

A ML research paper is a structured document where researchers present

- A new method or algorithm
- Results comparing methods on real-world data
- Analysis of why their approach works (or doesn't)

Typical structure

- Abstract: short summary of the main idea and results.
- Introduction: the problem, why it matters, and what the paper contributes.
- Related work: previous approaches and how this paper is different.
- Method/approach: the technique, model architecture, or algorithm.
- Experiments & results: how well the method works, often using real data.
- Discussion: interpret the results, talks about limitations.
- Conclusion & future work: summarizes findings and suggests what's next.

Reading ML research papers – Why?

Reading research papers is **key** to become a good ML engineer.

Ensuring **skillset** remains relevant by staying **up-to-date** with latest trends and methods.

Better **understanding** of methodology assumptions, use-cases, strengths and limitations.

Inspiration for **real-world solutions**.

World economic forum: skills development and ***lifelong learning*** improve the employability of workers.

Finding ML research papers

Determine the **topics of interest** e.g., time series forecasting, pose estimation, summarization.

Finding research papers

- PapersWithCode, arxiv-sanity, Google Scholar, etc.
- Leading conferences/journals e.g. NeurIPS, ICML, ICLR, JMLR.

Leverage **reviews and surveys**

- **Keywords, nomenclature** and **taxonomy**.
- Extensive reference list.

Ensure a paper is **relevant**: check number of citations, review abstracts.

Explore **references** of relevant papers.

Reading ML research papers – How?

1. Understand the context and gain intuition on the paper: **title, abstract, and conclusion.**
 - Note keywords, datasets, methods, etc. (use tools like Logseq, Notion)
 - Gain overview of the paper's scope and objectives.
2. Familiarize yourself with the paper content: **introduction and figures.**
 - Presentation of the problem, past works, research gaps and paper contribution.
 - Develop your first intuition on the paper.
 - Studying figures at this stage enables a more critical perspective.
3. Deep reading: **full paper** from abstract to conclusion but **skipping complex math or formulations.**
 - Note key insights and takeaways along with skipped parts.
4. Final pass: **come back** to skipped notions/parts, use external sources to facilitate understanding.

Danke!

Hochschule Luzern
Informatik, ABIZ
Dr. Ludovic Amruthalingam
Dozent Forschung

ludovic.amruthalingam@hslu.ch

